Lecture 1 Series

① If $\{a_n\}_{n=1}^{\infty}$ is a sequence, then $\sum_{n=1}^{\infty} a_n$ is a series.

Eg: $\{\frac{1}{n}\}_{n=1}^{\infty} \to$ seq & $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots \to$ series

② If $\{a_n\}$ is a seq. of +ve terms, then $\sum a_n$ is a series of +ve terms.

③ Now $\because a_n > 0$ and $S_1 = a_1$, $S_2 = a_1 + a_2 \Rightarrow S_1 < S_2$

$S_3 = a_1 + a_2 + a_3 \Rightarrow S_2 < S_3$

$\vdots$

$S_n = a_1 + a_2 + \dots a_n \Rightarrow S_{n-1} < S_n$

$\{S_n\}$ can be only cgt or div $\Leftarrow$ $\{S_n\}$ is monotonically ↑ng.

(*) oscillating ??? (Not possible)

if bdd above (cgt) (M.C.T. of sequences)

If unbdd above (div)

Result: 1) If $\{S_n\}$ is cgt then $\sum a_n$ is cgt.

2) If $\{S_n\}$ is div. then $\sum a_n$ is div.

Note: $\{S_n\}$ is called seq. of partial sums & when $S_n \xrightarrow{\text{conv.}} l$, l becomes the sum of $\sum a_n$ & limit of $\{S_n\}$.

Eg. ① $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ $\quad S_1 = a_1 = \frac{1}{1\cdot 2} = 1 - \frac{1}{2}$, $S_2 = a_1 + a_2 = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3}$

$= 1 - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \frac{1}{3}$

$= 1 - \frac{1}{3}$

|||ly $S_3 = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} = 1 - \frac{1}{4}$

$\therefore S_n = 1 - \frac{1}{n+1} = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5} \dots\right\}$

$\therefore S_n$ mono. ↑, is it bdd above ?? Yes $\because$ $\boxed{\sup = 1, \ \lim S_n = 1}$

(discussed in previous class)

$\therefore$ By Result 1)

$\Rightarrow \sum a_n$ is cgt $\left[\{S_n\} \text{ is cgt by M.C.T}\right]$

Note: This is an example of Telescopic series also. Formal def. could be: a series whose general term say a_n can be written in the form $a_n = t_{n+1} - t_n$ i.e. difference of two consecutive terms of some other sequence, say $\{t_n\}$.

Eg. ② $\sum n^2$ = sum of squares (Intermediate maths) S_n = sum of n squares $= \frac{n(n+1)(2n+1)}{6}$

& this is unbdd above. Hence by Result ② $\sum n^2 \to$ div.

A peculiar example : $\sum r^n$ conv. for $0 < r < 1$ & diverges for $r > 1$.

As of now, it was easy to compute partial sums & then applying Result ① & ②, we were able to discuss the conv. & div. of series

Now, let us look at ~~this~~ some bizarre examples $\sum a_n = \sum \frac{1}{\sqrt{5n^2+7n+11}}$

or $\sum \frac{1}{2n^2-n}$ or $\sum \frac{1}{n+5n}$.

Q. Can you still compute the partial sums? If yes, then how & how much time & effort it is going to take?

⇒ To make it easy, we will see some "convergence" tests. Before moving ahead, let me introduce you to the condition necessary for a sequence to converge.

Imp Remark: For $\sum a_n$ to converge, $\lim a_n$ should $\to 0$.

Caution! ULTA need not be true. i.e. If $\lim a_n = 0$, we can't say, $\sum a_n$ will definetly converge, it may or it may not.

But if $\lim a_n \not\to 0$ then for sure, it can't converge.

Test 1: The p-series test. $\sum \frac{1}{n^p}$ conv. $p > 1$, div. $p \le 1$

Eg. ① $\sum a_n = \sum n \sin \frac{1}{n} \Rightarrow a_n = n \sin 1/n = \frac{\sin 1/n}{1/n}$

$\lim a_n = 1 \neq 0 \therefore \sum a_n$ not conv. (necessary condition fail)

② $\sum a_n = \sum \frac{1}{n} \Rightarrow a_n = \frac{1}{n} \Rightarrow \lim a_n = 0$ (nec. cond. pass)

"may" converge ⇒ Test 1 apply ⇒ $p = 1 \therefore \sum a_n$ div. (p series test)

③ $\sum a_n = \sum \frac{1}{n^{3/2}} \Rightarrow a_n = \frac{1}{n^{3/2}}$ $\lim a_n = 0$

& $\because p = 3/2 > 1 \therefore \sum a_n$ conv. by p-series test.